

SECTION I DIFFERENTIATION

Some of the answers are simplified. We need the simplified version when we want to sketch graphs and in the textbook the answers are mostly simplified.

You DO NOT have to simplify your answers in this exercise.

1. $8x^3 - 6x + 5$

2. $\frac{-1}{2}x^{\frac{-3}{2}} + \frac{1}{2}x^{\frac{-1}{2}} = \frac{x-1}{2x\sqrt{x}}$

3. $6x + (-2)\frac{1}{3}x^{-3} + 1 = \frac{18x^4 - 2 + 3x^3}{3x^3}$

4. $2x - 3\left(\frac{7}{3}\right)x^{\frac{4}{3}} = 2x - 7x^{\frac{4}{3}}$

5. $1 + 0 - 2t^{-3} = \frac{t^3 - 2}{t^3}$

6. $\frac{1}{2}x^{\frac{-1}{2}} - \left(-\frac{1}{2}\right)x^{\frac{-3}{2}} = \frac{x+1}{2x\sqrt{x}}$

7. $12(-1)x^{-2} - 4(-3)x^{-4} + (-4)x^{-5} = \frac{-4(3x^3 - 3x + 1)}{x^5}$

8. $\frac{1}{3}x^{\frac{-2}{3}} - \frac{3}{x^4}$

9. $4x^3 - 8(3)x^2 + 2(2)x - 1 + 0 = 4x^3 - 24x^2 + 4x - 1$

10. $7(3)y^2 - 5(2)y + 3 + 0 = 21y^2 - 10y + 3$

11. $9(-3)x^{-4} + 2(-2)x^{-3} + 0 = -\frac{27+4x}{x^4}$

12. $-2(4)x^3 + (-2)x^{-3} - 3\left(\frac{3}{4}\right)x^{\frac{-1}{4}} = -\frac{32x^{\frac{25}{4}} + 8\sqrt[4]{x} + 9x^3}{4x^{\frac{13}{4}}}$

13. $12(4)t^3 + 3(3)t^2 + 5(-2)t^{-3} - 0 = \frac{48t^6 + 9t^5 - 10}{t^3}$

14. $4(-2)x^{-3} - 7\left(\frac{1}{2}\right)x^{\frac{-1}{2}} + 8(3)x^2 + 0 = -\frac{16\sqrt{x} + 7x^3 - 48x^{\frac{11}{2}}}{2x^{\frac{7}{2}}}$

15. $3(3)x^2 = 2(2)x - 1 + 0 = 9x^2 + 4x - 1$

16. $4(3)x^2 - 7(2)x + 8 - 0 = 12x^2 - 14x + 8$

17. $3x^2 - 3 - 2(-4)x^{-5} = 3x^2 - 3 + 8x^{-5}$

18. $1 + 0 + 4(-1)x^{-2} = 1 - \frac{4}{x^2}$

19. $-3(-3)u^{-4} + 4(2)u + (-2)u^{-3} = 9u^{-4} + 8u + \frac{-2}{u^3}$

20. $\frac{d}{dx}\left(x^{\frac{3}{2}} + x^{\frac{-5}{2}}\right) = \frac{3x^{\frac{1}{2}} - 5}{2x^{\frac{7}{2}}}$

21. $5(x^3 + 7x^2 + 5)^4(3x^2 + 7(2)x + 0) = 5(x^3 + 7x^2 + 5)^4(3x^2 + 14x)$

22. $\frac{1}{2}(x^4 + x^2 + 2)^{\frac{-1}{2}}(4x^3 + 2x + 0) = \frac{1}{2}(x^4 + x^2 + 2)^{\frac{-1}{2}}(4x^3 + 2x)$
23. $\frac{1}{2}(x + 1)^{\frac{-1}{2}}(x^2 + 1) + \sqrt{x + 1})(2x) = \frac{5x^2 + 4x + 1}{2\sqrt{1 + x}}$
24. $\frac{1}{2}(t^2 + 2t + 3)^{\frac{-1}{2}}(2t + 2) = \frac{t + 1}{\sqrt{t^2 + 2t + 3}}$
25. $2x(x^3 - 1) + x^2(3x^2) = 5x^4 - 2x$
26. $4(x^3 - 3x)^3(3x^2 - 3) = 12x(x^2 - 3)(x^2 - 1)$
27. $\sqrt{1 - x^2} + (x)\frac{1}{2}(1 - x^2)^{\frac{-1}{2}}(-2x) = \frac{1 - 2x^2}{\sqrt{1 - x^2}}$
28. $(2)(3x^2 + 2) + (2x - 4)(6x) = 18x^2 - 24x + 4$
29. $[3(2x - 3)^2(2)](4x + 2)^2 + (2x - 3)^3[(2)(4)(4x + 2)]$
30. $[2(7x + 3)(7)](3x^2 - 14x + 5) + (7x + 3)^2(3(2)x - 14)$
31. $(24y^2)(3y^2 - 5y + 10)^2 + (8y^3 - 2)[2(3y^2 - 5y + 10)(6y - 5)]$
32. $\frac{2}{3}(2x^3 - 3)^{\frac{-1}{3}}(6x^2)$
33. $\frac{1}{2}(3x^2 - 2x + 1)^{\frac{-1}{2}}(6x - 2) = \frac{3x - 1}{\sqrt{3x^2 - 2x + 1}}$
34. $2(3x + 4) + (2x - 3)(3) = 12x - 1$
35. $(6x^2)(x^4 + x) + (2x^3 - 1)(4x^3 + 1) = 14x^6 + 4x^3 - 1$
36. $(\frac{-1}{x^2})(x^2 - 5) + (\frac{1}{x} + 3)(2x) = \frac{6x^3 + x^2 + 5}{x^2}$
37. $4(2x + 1)(x^2 + 2)^3 + (2x + 1)^2 3(x^2 + 2)^2(2x) = 2(2x + 1)(x^2 + 2)^2[8x^2 + 3x + 2]$
38. $\frac{1}{2}(3x^2 + 1)^{\frac{-1}{2}}(6x) = \frac{3x}{\sqrt{3x^2 + 1}}$
39. $6(2t - 5)^2$
40. $-3(6)(6x - 5)^{-4} = -\frac{18}{(6x - 5)^4}$
41. $\frac{(2x+1)(x^2+1)-(x^2+x+1)2x}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$
42. $\frac{2(3x+2)-(2x+3)(3)}{(3x+2)^2} = \frac{-5}{(3x+2)^2}$
43. $\frac{2(3u-5)-(2u+1)(3)}{(3u-5)^2} = -\frac{13}{(3u-5)^2}$
44. $\frac{(2x+5)x^2-(x^2+5x-1)(2x)}{(x^2)^2} = \frac{2-5x}{x^3}$
45. $\frac{d}{dx} \left(\frac{(x^2 + 3x + 2)}{(x^2 - 3x + 2)} \right) = \frac{(2x + 3)(x^2 - 3x + 2) - (x^2 + 3x + 2)(2x - 3)}{(x^2 - 3x + 2)^2} = \frac{-6(x^2 - 2)}{(x - 1)^2(x - 2)^2}$

46. $\frac{1}{2} \left(\frac{x^2+1}{x^2+4} \right)^{-\frac{1}{2}} \left(\frac{(2x)(x^2+4) - (x^2+1)2x}{(x^2+4)^2} \right) = \frac{3x}{\sqrt{x^2+1} \sqrt{x^2+4} (x^2+4)^2}$
47. $\frac{1}{2\sqrt{t}} (1 + 5t^2) = \frac{(1 + 3t^2) \sqrt{t} - (t + t^3) \frac{1}{2} t^{-\frac{1}{2}}}{(\sqrt{t})^2} = \frac{1 + 5t^2}{2\sqrt{t}}$
48. $2 \left(\frac{x+1}{x-1} \right) \left(\frac{(x-1) - (x+1)}{(x-1)^2} \right) = - \frac{-4(x+1)}{(x-1)^3}$
49. $\frac{d}{dx} (10(\sqrt{x} + 4)^{-1}) = -10(\sqrt{x} + 4)^{-2} \left(\frac{1}{2} x^{-\frac{1}{2}} \right) = \frac{-5}{(\sqrt{x} + 4)^2 \sqrt{x}}$
50. $\frac{d}{dx} (8(4 + x^2)^{-1}) = -8(4 + x^2)^{-2} (2x) = - \frac{16x}{(4 + x^2)^2}$
51. $\frac{4(y^2 + 1) - 4y(2y)}{(y^2 + 1)^2} = \frac{4(1 - y^2)}{(y^2 + 1)^2}$
52. $\frac{d}{dx} \left(3(2x + 1)^{-\frac{1}{2}} \right) = 3 \left(-\frac{1}{2} \right) (2x + 1)^{-\frac{3}{2}} (2) = - \frac{3}{(2x + 1)^{\frac{3}{2}}}$
53. $\frac{(6x - 2)(4x^2 - 5) - (3x^2 - 2x + 3)(8x)}{(4x^2 - 5)^2} = \frac{2(4x^2 - 27x + 5)}{(4x^2 - 5)^2}$
54. $\frac{[(2)(3x + 5) + (2x - 4)(3)](2x^2 + 7) - (2x - 4)(3x + 5)(4x)}{(2x^2 + 7)^2} = \frac{2(82x + 2x^2 - 7)}{(2x^2 + 7)^2}$
55. $\frac{2(x^2 + 2x) - (2x - 3)(2x + 2)}{(x^2 + 2x)^2} = \frac{-2(x^2 - 3x - 3)}{x^2(x + 2)^2}$
56. $\frac{(-2 + 6t)(t^2 + 2) - (4 - 2t + 3t^2)2t}{(t^2 + 2)^2} = \frac{2(t^2 + 2t - 2)}{(t^2 + 2)^2}$
57. $6x^2 + \frac{-1(x^3) - (2-x)(3x^2)}{x^6} = \frac{2(3x^6 + x - 3)}{x^4}$
58. $\frac{1(x+1) - (x-1)1}{(x+1)^2} = \frac{2}{(x+1)^2}$
59. $\frac{2x(x^2 + 1) - x^2(2x)}{(x^2 + 1)^2} = \frac{2x}{(x^2 + 1)^2}$
60. $\frac{d}{dt} (t^4 - 2t + 1)^{-1} = -1(t^4 - 2t + 1)^{-2} (4t^3 - 2) = \frac{-2(2t^3 - 1)}{(t^4 - 2t + 1)^2}$
61. $\frac{d}{dx} (\sin(x^2 + 2)) = [\cos(x^2 + 2)](2x) = 2x \cos(x^2 + 2)$
62. $(\cos(3x))(3) = 3 \cos 3x$
63. $(-\sin(3y))(3) = -3 \sin 3y$
64. $[-\sin(x^4 + 7)]4x = -4x^3 \sin(x^4 + 7)$
65. $[\sec^2(\sin x)] \cos x = \cos x \times \sec^2(\sin x)$
66. $(\sec^2(t^2 + 5))(2t) = \frac{2t}{\cos^2(t^2 + 5)}$

67. $[\cos(3x + 2)](3) = 3 \cos(3x + 2)$
68. $2 \cos x - \sec^2 x = \frac{2 \cos^3 x - 1}{\cos^2 x}$
69. $= 1 + \sin x$
70. $\frac{(\cos x)x - \sin x(1)}{x^2} = \frac{x \cos x - \sin x}{x^2}$
71. $[\sec^2(2x - x^3)](2 - 3x^2) = \frac{2 - 3x^2}{\cos^2(-2x + x^3)}$
72. $(\cos(3\theta^2 - 2\theta + 1))(6\theta - 2) = (6\theta - 2) \cos(3\theta^2 - 2\theta + 1)$
73. $3[\cos(x^2)](2x) + 2 \cos x - 0 = 6x \cos x^2 + 2 \cos x$
74. $3 \cos x + 2 \sin x$
75. $(-\sin(\pi x - 1))(\pi) = -\pi \sin(\pi x - 1)$
76. $3[-\sin(3x)] - 3 \sec^2(3x) = \frac{-3(\sin 3x \cos^2 3x + 1)}{\cos^2 3x}$
77. $2x \sin x + x^2 \cos x = x(2 \sin x + x \cos x)$
78. $[-\sin(\tan x)] \sec^2 x = \frac{-\sin(\tan x)}{\cos^2 x}$
79. $2 \cos(2x + 3)$
80. $(\cos 2\pi t)(2\pi) = 2\pi(\cos 2\pi t)$
81. $\left(\cos \frac{1}{\sqrt{x}}\right)\left(-\frac{1}{2}x^{-\frac{3}{2}}\right) = -\frac{\cos \frac{1}{\sqrt{x}}}{2x^{\frac{3}{2}}}$
82. $\left(\cos \frac{1}{x}\right)\left(-\frac{1}{x^2}\right) = -\frac{\cos \frac{1}{x}}{x^2}$
83. $(-\sin(-x))(-1) = -\sin x$
84. $\left(\sec^2 \pi\left(\frac{1}{2} - x\right)\right)(-\pi)$
85. $2x \sec^2(x^2 + 1)$
86. $5 \sec^2 5t$
87. $\cos x + x(-\sin x) = \cos x - x \sin x$
88. $\frac{(\cos x)(1 + x) - \sin x}{(1 + x)^2}$
89. $4(2)(\sin 3x)(\cos 3x)(3) = 12 \sin 6x$
90. $\frac{(-\sin x) \sin x - \cos x(\cos x)}{(\sin x)^2} = \frac{-1}{(\sin x)^2}$
91. $2x \sin x + (x^2 + 3) \cos x$
92. $\frac{1}{2}(\sin x)^{-\frac{1}{2}} \cos x = \frac{\cos x}{2\sqrt{\sin x}}$

93. $2(\cos(x^3))(-\sin(x^3))(3x^2) = -3x^2(\sin 2x^3)$
94. $\cos 5x + x(-\sin 5x)(5) = \cos 5x - 5x \sin 5x$
95. $2 \cos(2y) \times \cos(3y) + \sin(2y)(3(-\sin(3y))) = 2 \cos 2y \cos 3y - 3 \sin 2y \sin 3y$
96. $2(\sin(3x))(3) \cos(3x) + \cos(5x)(2)(-\sin(5x))(5) = 3 \sin 6x - 5 \sin 10x$
97. $2x \tan \frac{1}{x} + x^2(\sec^2(\frac{1}{x}))\left[\frac{-1}{x^2}\right] = 2x \tan \frac{1}{x} - \sec^2(\frac{1}{x})$
98. $-1(\sin x - x \cos x)^{-2}[\cos x - (\cos x + x(-\sin x))] = \frac{-x \sin x}{(\sin x - x \cos x)^2}$
99. $\frac{(\cos x - \sin x)(\tan x) - (\sin x + \cos x) \sec^2 x}{(\tan x)^2}$
100. $\frac{d}{dx}(1) = 0$
101. $\frac{2x \tan x - x^2(\sec^2 x)}{\tan^2 x}$
102. $\frac{1}{2}(\cos(2x))^{\frac{-1}{2}}(-\sin(2x))(2) = \frac{-\sin 2x}{\sqrt{\cos 2x}}$
103. $2 \sin(x - 3) \cos(x - 3) = \sin(2(x - 3))$
104. $3 \cos^2(t^2 - t)[- \sin(t^2 - t)](2t - 1) = -3(2t - 1) \cos^2(t^2 - t) \sin(t^2 - t)$
105. $\cos x \cos^2 x + (\sin x)2(\cos x)(-\sin x) = \cos x(\cos^2 x - 2 \sin^2 x)$
106. $\frac{1}{3} \left[\sin^{\frac{-2}{3}}(2x) \right] [\cos(2x)](2) = \frac{2 \cos 2x}{3 \sin^{\frac{2}{3}} 2x}$
107. $[7 \sin^6(2 - 4x)](\cos(2 - 4x))(-4) = -28 \sin^6(2 - 4x) \cos(2 - 4x)$
108. $2 \cos x + 6 \sin x \cos x = 2 \cos x(1 + 3 \sin x)$
109. $3 \cos^2 u(-\sin u) + 4(2) \cos u(-\sin u) = -\sin u \cos u(3 \cos u + 8)$
110. $5(2)[\cos(x + 2)](-\sin(x + 2)) + 3(-\sin(x + 2)) = -\sin(x + 2)(10 \cos(x + 2) + 3)$
111. $10x - 6 \tan x \sec^2 x + \cos x = \frac{10x \cos^3 x - 6 \sin x + \cos^4 x}{\cos^3 x}$
112. $\frac{(\sec^2 x)(1 - \tan x) - (\tan x)(-\sec^2 x)}{(1 - \tan x)^2}$
113. $\frac{3}{2} \sin^{\frac{1}{2}}(x + 1) \cos(x + 1) = \frac{3}{2} \sin^{\frac{1}{2}}(x + 1) \cos(x + 1)$
114. $\frac{d}{dx}(1) = 0$
115. $(\cos \sqrt{x})^{\frac{1}{2}} x^{\frac{-1}{2}} + \frac{1}{2}(\sin x)^{\frac{-1}{2}} \cos x = \frac{(\cos \sqrt{x})}{2\sqrt{x}} + \frac{\cos x}{2\sqrt{\sin x}}$
116. $(\cos x)(\sin x + \cos x) + (\sin x)(\cos x - \sin x) = \sin 2x + \cos 2x$
117. $\frac{d}{dx}(\sin 2x) = 2 \cos 2x = 2 \cos 2x$
118. $\frac{d}{du}(\sin u)^{-1} = (-1)(\sin u)^{-2} \cos u$
119. $2(\cos x)(-\sin x)(\sin x) + \cos^2 x \cos x = 3 \cos^3 x - 2 \cos x$

120. $\frac{d}{dx} \sqrt{\cos^2 x} = \frac{1}{2} (\cos^2 x)^{-\frac{1}{2}} - 2 \sin x \cos x$
121. $1 + 2 \tan x \sec^2 x$
122. $\frac{\cos x + x \sin x}{\cos^2 x}$
123. $\sec^2 x \cos x + \tan x(-\sin x) = \cos x$
124. $\cos t \cos t + \sin t(-\sin t) = \cos^2 t - \sin^2 t = \cos 2t$
125. $2xe^{x^2}$
126. $5e^{5x}$
127. $(2x + 3)e^x + (x^2 + 3x)e^x = e^x(x^2 + 5x + 3)$
128. $e^x + xe^x - (-1)e^{-x} = \frac{e^{2x} + xe^{2x} + 1}{e^x}$
129. $\frac{d}{dx} (e^{x^2+x+1}) = (2x + 1)e^{x^2+x+1} = (2x + 1)e^{x^2+x+1}$
130. $\frac{d}{dx} (e^{x^2-x+1}) = (2x - 1)e^{x^2-x+1} = (2x - 1)e^{x^2-x+1}$
131. $-e^{-u}$
132. $\frac{2}{3} e^{\frac{2x}{3}}$
133. $e^{\sqrt{x}} \frac{1}{2} x^{-\frac{1}{2}} = \frac{e^{\sqrt{x}}}{2\sqrt{x}}$
134. $3e^{3x} + 2(2)e^{2x} - 3e^x = 3e^{3x} + 4e^{2x} - 3e^x$
135. $2xe^{x^2-2}$
136. $\frac{2e^{2x}(2 - e^{2x}) - (1 + e^{2x})(-2e^{2x})}{(2 - e^{2x})^2} = \frac{6e^{2x}}{(2 - e^{2x})^2}$
137. $3e^{3x-1} - 4(-1)e^{-x} = \frac{3e^{4x-1} + 4}{e^x}$
138. $(-\sin(e^t))e^t = -e^t \sin(e^t)$
139. $3(2)e^{2x} - 4e^x = 2e^x(3e^x - 2)$
140. $e^{3\cos 2x} \times 3(-\sin 2x)(2) = -6(\sin 2x)e^{3\cos 2x}$
141. $-2e^{-2x} + 4(-3)e^{-3x} = -2e^{-2x} - 12e^{-3x}$
142. $2e^{2x+1}$
143. $\frac{1}{2}(2)e^{2x} = e^{2x}$
144. $(\cos x)e^{\sin x}$
145. $2e^{2x}$
146. $2e^t + 2te^t = 2e^t(1 + t)$
147. $\frac{d}{dx} (1 - e^{-x})^{-1} = -1(1 - e^{-x})^{-2}(-(-1)e^{-x}) = \frac{-e^{-x}}{(1 - e^{-x})^2}$
148. $\frac{-e^{-x}x - e^{-x}(1)}{x^2} = \frac{-e^{-x}(x + 1)}{x^2}$

149. $2xe^{-x} + x^2(-1)e^{-x} = \frac{x(2-x)}{e^x}$
150. $e^{-\frac{1}{x^2}}(-1(-2)x^{-3}) = \frac{2}{x^3}e^{-\frac{1}{x^2}}$
151. $e^{\sqrt{x^2+1}}\left(\frac{1}{2}(x^2+1)(2x)\right) = \frac{xe^{\sqrt{x^2+1}}}{\sqrt{x^2+1}}$
152. $\frac{e^x+e^{-x}}{2}$
153. $2e^{2x}\cos 3x + e^{2x}(3)(-\sin 3x) = e^{2x}(2\cos 3x - 3\sin 3x)$
154. $4(-\sin 4x)e^{\cos 4x} = -4\sin 4xe^{\cos 4x}$
155. $(2x)2^x + x^2(2^x \ln 2) = 2^x(2x + (\ln 2)x^2)$
156. $3^{5x}5 \ln 3 = (5 \ln 3)3^{5x}$
157. $4x^3 + 4^x \ln 4 = 4x^3 + 4^x \ln 4$
158. $9^{-x}(-1) \ln 9 = \frac{-\ln 9}{9^x}$
159. $(\sec^2 5^x)(5^x \ln 5) = \frac{5^x \ln 5}{\cos^2 5^x}$
160. $3^{4x+1}(4 \ln 3) + 2^{4x+2}(4 \ln 2) = 4[(\ln 3)3^{4x+1} + (\ln 2)2^{4x+2}]$
161. $3^{y^2+1}(2y \ln 3) = (2y \ln 3)3^{y^2+1}$
162. $2^x \ln 2 = (\ln 2)2^x$
163. $2^{-x}(-1) \ln 2 = \frac{-\ln 2}{2^x}$
164. $\left(\frac{1}{2}\right)^t \ln\left(\frac{1}{2}\right) = -\ln 2\left(\frac{1}{2}\right)^t$
165. $e^x \ln x + e^x\left(\frac{1}{x}\right) = e^x\left(\ln x + \frac{1}{x}\right)$
166. $\frac{\cos x}{\sin x} = \cot x$
167. $\frac{d}{dx}(\ln x)^{-1} = -1(\ln x)^{-2}\left(\frac{1}{x}\right) = \frac{-1}{x(\ln^2 x)}$
168. $\frac{d}{dx}(\ln 3x + \ln e^{-x}) = \frac{d}{dx}(\ln 3x - x) = \frac{3}{3x} - 1 = \frac{1}{x} - 1$
169. $\frac{d}{dx}(\ln(x-1) - \ln(x^2+1)) = \frac{1}{x-1} - \frac{2x}{x^2+1} = \frac{-(x^2-2x-1)}{(x-1)(x^2+1)}$
170. $\frac{d}{dx}(\ln e^x - \ln(1+e^x)) = \frac{d}{dx}(x - \ln(1+e^x)) = 1 - \frac{e^x}{1+e^x} = \frac{1}{1+e^x}$
171. $\frac{d}{dx}(\sin 2x) = 2 \cos 2x$
172. $\frac{d}{dx}\left(\frac{1}{2}[\ln x - \ln(x^2+1)]\right) = \frac{1}{2}\left[\frac{1}{x} - \frac{2x}{x^2+1}\right] = \frac{1-x^2}{2x(x^2+1)}$
173. $\frac{d}{du}(2 \ln u) = \frac{2}{u}$
174. $\frac{d}{dx}(\ln 10 - \ln x) = -\frac{1}{x}$
175. $\frac{d}{dx}(x \ln 10) = \ln 10$

$$176. \frac{3}{3x} + \frac{4}{x} = \frac{5}{x}$$

$$177. 2x \ln 2x + x^2 \left(\frac{2}{2x} \right) = x(\ln 4 + 2 \ln x + 1)$$

$$178. \frac{d}{dx}(-\ln x) = -\frac{1}{x}$$

$$179. \ln x + x \left(\frac{1}{x} \right) = \ln x + 1$$

$$180. -\frac{1}{x}$$

$$181. 3(\ln y)^2 \frac{1}{y} = \frac{3 \ln^2 y}{y}$$

$$182. \ln \sqrt{x} + x \frac{d}{dx} \left(\frac{1}{2} \ln x \right) = \ln \sqrt{x} + x \left(\frac{1}{2} \right) \left(\frac{1}{x} \right) = \ln \sqrt{x} + \frac{1}{2}$$

$$183. \frac{7}{7x}$$

$$184. \frac{1}{2} (\ln x)^{\frac{-1}{2}} \left(\frac{1}{x} \right) = \frac{1}{2x\sqrt{\ln x}}$$